



The conditioning of surprisal and many-to-one mappings in speech variation: Articulatory efficiency and perceptual recoverability

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Speech variation under information-theoretic constraints

- Different information-theoretic (Shannon 1948) correlates are shown to predict different types of variation:
- More informative/unpredictable items maintain higher distinctiveness/are more resilient to variation.
- For example:
 - Frequencies correlate w/ word durations (Baker & Bradlow 2009) & vowel merger rates (Kang, Yoon & Han 2015).
 - Functional load (# of minimal pairs) correlates w/ merger rates & vowel lengths (Wedel *et al.* 2013).